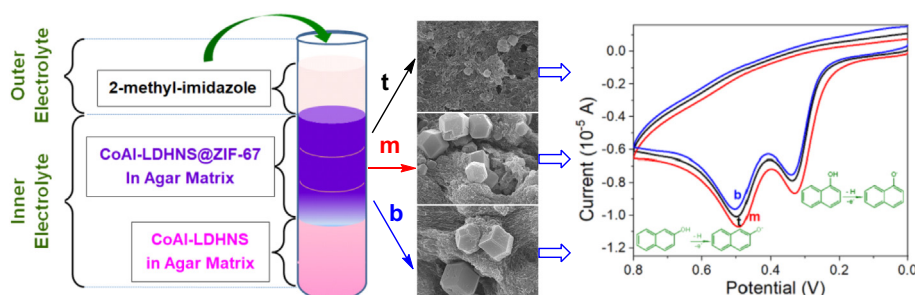


Regular Article

Electrochemical sensing for naphthol isomers based on the in situ growth of zeolitic imidazole framework-67 on ultrathin CoAl layered double hydroxide nanosheets by a reaction–diffusion technique

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GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 6 January 2021

Revised 28 April 2021

Accepted 29 April 2021

Available online 4 May 2021

Keywords:

Zeolitic imidazole frameworks
Ultrathin layered double hydroxide nanosheets
Reaction–diffusion framework
Naphthol isomers sensor
Differential pulse voltammetry

ABSTRACT

It is established that ultrathin layered double hydroxide nanosheets (LDHNS) and zeolitic imidazole frameworks (ZIF) are desirable electrochemical sensing modifiers owing to their large surface area and abundant catalytic sites. Integration of them is thus an effective solution to maximize their electrocatalytic activity. Herein, a novel reaction–diffusion framework (RDF) technique is applied for the in situ growth of ZIF-67 on ultrathin CoAl-LDHNS (CoAl-LDHNS@ZIF-67). In a confined space of the agar gel matrix of RDF, the coordination reaction between organic ligands and CoAl-LDHNS without an additional Co²⁺ source achieves the controllable growth of ZIF-67 crystals through a long vertical diffusion. The prepared composite comprises both CoAl-LDHNS and ZIF-67 components with a certain ratio and provides a large surface area and amply catalytic sites, thus realizing a rapid transfer of electron and mass. The CoAl-LDHNS@ZIF-67 modified electrode is employed for the simultaneous detection of naphthol isomers by differential pulse voltammetry. Naphthol isomers display anodic reactions with a wide peak potential difference, allowing their simultaneous detection feasible. Voltammetric responses of α-naphthol and β-naphthol follow good linearity against the concentration in a wide range from 0.3 to 150 μM with limits of detection of 54 and 82 nM, respectively. The proposed sensor also demonstrates excellent selectivity, stability, reproducibility, and practicability for the simultaneous detection of naphthol isomers.

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1. Introduction

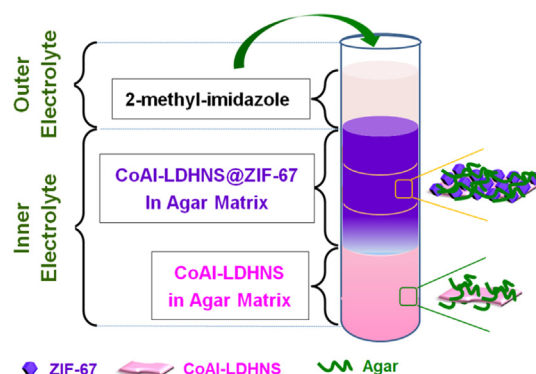
Naphthol (NAP) is a class of important raw materials and intermediates in the organic synthesis and production of diverse chemicals including dyes, greases, rubbers, insecticides, antiseptic substances, and drugs [1–3]. NAP has two isomers of α -naphthol (α -NAP) and β -naphthol (β -NAP). The former can destroy the male reproductive system, whereas the latter can result in serious systemic intoxication owing to its notorious chemical teratogenesis [1,4–7]. NAP isomers are also metabolites of naphthalene, which has been listed as an endocrine disruptor with high mutagenicity, teratogenicity, and carcinogenicity [4,8]. Because of the extensive applications, NAP compounds are inevitably discharged into the soil and waterbody as industrial effluent, thus leading to serious environmental pollution. Studies have shown that α -NAP and β -NAP are non-degradable and toxic pollutants with median lethal dose (LD_{50}) values of 1700–300 mg kg^{−1} and 4200 mg kg^{−1}, respectively [9]. Hence, quantitation of NAP isomers becomes more and more significant in public safety and environmental assessment. The presence of phenolic hydroxyl endows NAP molecules with intrinsic electrochemical activity, which makes it possible to explore an efficient electrochemical technique for fast and sensitive detection of NAP isomers. Among many techniques, the electroanalytical method has demonstrated several merits such as simple operation, low cost, high sensitivity, and fast on-the-spot analysis. However, both NAP isomers frequently co-occur in the same environment sample owing to their similar structures and physicochemical properties, which could bring out the mutual electrochemical interference in some cases [10]. Consequently, only a few electrochemical methods were developed for the simultaneous detection of NAP isomers owing to their adjacent anodic peak potentials and easy electropolymerization [2,3].

It is imperative but still challenging to explore high-efficient sensing material for the construction of high-sensitive and high-selective analytical methods for simultaneous detection of NAP isomers. Metal-organic framework (MOF) has emerged as a kind of supermolecular materials composed of metal ions with organic linkers, and MOF-based compounds have shown promising applications in electrochemical sensing due to their tunable structure, high porosity, and abundant active sites [11–13]. Zeolitic imidazole framework (ZIF), a subclass of MOF, consists of inorganic metallic ions or clusters and organic imidazolate linkers. Of the MOF family, ZIF possesses higher porosity, diverse structures, better thermal and chemical stability, and ease of synthesis [14–16]. As a consequence, ZIF and their derivatives have been widely used in gas absorption [17], molecular recognition [18], energy storage [19], electrocatalysis [20], and electrochemical sensing [10,21,22], etc. Nevertheless, bare ZIF usually exhibits poor dispersibility and stability in an electrolyte solution as well as inferior electrocatalytic activity. The introduction of other supporting matrices would be helpful for the in situ growth of ZIF crystals to improve their adsorptive and electrocatalytic performance [23,24]. Impressively, inorganic layered materials have demonstrated great potential in the in situ crystallization of ZIF without impairing their large surface area [25,26].

Layered double hydroxides (LDH) are a class of 2-dimensional (2D) inorganic anionic materials with a generic formula of $[M_1^{II}_x M_2^{III}_{1-x}(\text{OH})_2]^{x+}[(A^{n-})_{x/n} \cdot m\text{H}_2\text{O}]$. M^{II} and M^{III} metal ions locate at the centers of octahedral MO_6 cells, whereas OH^- anions occupy the vertexes. These MO_6 units extend to form the 2D brucite-like layer by sharing their edges. The isomorphous replacement of M^{II} by M^{III} yields the forever positively charged brucite-like layers, accompanying the charge-balancing interlamellar anions (A^{n-}) and water molecules [27]. Thanks to the specific lamellar structure of LDH, their surface unsaturated metal atoms could act as reactive

sites for the direct growth of ZIF nanocrystals [28]. Recently, the growth of ZIF-8 on ZnAl-LDH surface as adsorbents without adding an additional zinc source has been reported [26,29]. Besides, the in situ decorated CoAl-LDH with ZIF-67 was prepared as the precursor of electrocatalyst for supercapacitors by a hydrothermal method [25]. In the above studies, all LDH supporting materials employed are in the bulk-form, which cannot completely release reactive and catalytic sites to achieve a full performance since a great many of MO_6 active sites are embedded in stacked layers [30,31]. Especially for the in situ growth of ZIF on the LDH matrix [25,26,29], it involved an additional Co ion source, which thus gave rise to the random growth of ZIF-67 without controllable sizes and morphologies. Excitingly, our recent work found that ultrathin LDH nanosheets (LDHNS) can serve as a more desirable template for the in situ preparation of ZIF crystals by offering much more metallic reactive centers as the only Co^{2+} source than the bulk-LDH. However, it seems that most ultrathin LDHNS were converted into ZIF-67 crystals in this unrestricted reaction system and the LDHNS component only plays a limited role in electrocatalysis [2]. Furthermore, the growth of ZIF crystals on LDH supports in this unrestrained reaction environment usually results in uncontrollable morphology and size [24,26], which are significant factors affecting the performance of catalysts. Many efforts have been focused on the morphology- and size-controllable synthesis of ZIF crystals by using reverse microemulsion [32], microwave [33], and reaction – diffusion framework (RDF) technique. In the RDF, an agar gel matrix is utilized to fix the precipitating metal cations (Zn^{2+} and/or Co^{2+} : inner electrolytes) and the ligand solution (2-methyl imidazole: outer electrolyte) then diffuses from the top of the gel. In this natural diffusion process, the coordination reaction takes place in the gel matrix, and the growth of ZIF crystals is thus be controlled by restraining the fast precipitation kinetics [14]. Our newest study further has demonstrated that the RDF technique plays a crucial role in the controllable preparation of ZIF-8 on graphene oxide (GO) in a vertical gradient. However, in this manner, the random fixations of Zn^{2+} on the surface of GO resulted in the unordered growth of ZIF-8 crystals in a poor polyhedral morphology with a wide size distribution [10].

Herein, ultrathin CoAl-LDHNS was firstly prepared in a formamide aqueous solution. Then we use the RDF technique for the direct growth of ZIF-67 on ultrathin CoAl-LDHNS. As shown in Scheme 1, the 2-methyl imidazole solution acting as an outer reactant was added on the top of the inner reactant of ultrathin CoAl-LDHNS in the agar gel matrix. Owing to the ultrathin nature, CoAl-LDHNS can release a great deal of highly reactive Co^{2+} sites to coordinate with imidazole ligands. Besides the only Co^{2+} source, CoAl-LDHNS can also serve as the substrate and a template for the in situ ordered growth of ZIF-67 units. The long reaction–diffu-



Scheme 1. The schematic diagram for the synthesis of CoAl-LDHNS@ZIF-67 through the RDF method.

sion can ensure the controllable preparation of ZIF-67 crystals on LDHNS substrate (LDHNS@ZIF-67) in a vertical gradient. The RDF technique in this work can realize the morphology- and size-controllable preparation of LDHNS@ZIF-67 consisting of two phases in a certain ratio. Consequently, LDHNS@ZIF-67 can provide more adsorptive and electrocatalytic sites toward NAP isomers by integrating the organic ligand moiety of ZIF and inorganic LDHNS. The LDHNS@ZIF-67 modified electrode thus displays excellent catalytic activity toward the oxidation of α -NAP and β -NAP.

2. Experimental

2.1. Chemicals

$\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (AR, 99.0%), $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (AR, 99.0%), formamide (AR, 99.0%), *N,N*-dimethylformamide (DMF) (AR, 99.8%), 2-methylimidazole (98.0%), agar (AR, 98.5%), α -naphthol (α -NAP, MW = 146.17) (AR, 99.0%), β -naphthol (β -NAP) (AR, 99.0%) were purchased from Aladdin Chemistry Co. (Shanghai, China). Other reagents were analytical grade and used without further purification. Phosphate buffer (PB, 0.1 M) was prepared by mixing 0.1 M of 0.1 M NaH_2PO_4 and 0.1 M Na_2HPO_4 . The pH values from 6 to 8 were adjusted by adding NaOH or HCl solution. The α -NAP and β -NAP stock solution (50 mM) were respectively prepared by dissolving 0.7208 g of α -NAP or β -NAP in 100 mL ultrapure water for storage at 4 °C. The stock solutions with different concentrations were prepared by diluting with PB solution.

2.2. Synthesis of CoAl-LDHNS

According to the previous procedure [34], $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (2 mmol) and $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (1 mmol) were dissolved into the mixed solvent of formamide and water (*v:v* = 5:3) with their final concentrations as 40 and 20 mM, respectively. Then a 0.7 M NaOH solution was added dropwise to the above mixture within 3 h under stirring to maintain a pH of 8.5. After aging for 30 min at room temperature, the mixture was subjected to centrifugation for 5 min at 4000 rpm to remove bulk LDH. Then the pink CoAl-LDHNS colloid was collected through centrifuging for 10 min at 10000 rpm. The solid content is determined to be around 10%.

2.3. Synthesis of CoAl-LDHNS@ZIF-67

640 mg obtained LDHNS colloid was added to a round-bottom flask containing 8 mL ultrapure water and 8 mL *N,N*-dimethylformamide (DMF). After stirring for 15 min at room temperature, 80 mg agar was added to and the mixture was heated to boil and stirred at least 15 min to ensure that the agar was dissolved. The resultant solution was rapidly and discreetly poured into a series of plastic tubes (height: 10 cm; diameter: 1 cm) to fill the two-thirds height of each tube and these tubes were then sealed with parafilm and allowed to gelatinize for 2 h. After complete gelation of the inner portion, the outer solution (3600 mg 2-methylimidazole was dissolved in 16 mL of the 1:1 mixture of water/DMF) was poured into the top of those tubes. Next, these tubes were covered and allowed for another 72 h to undergo the reaction – diffusion and the formation of ZIF-67 crystals on CoAl-LDHNS. Afterward, the top 3 mm content of each tube was discarded to exclude the solid particles from the leading white band. The rest content of each tube was divided into three equidistant sections of about 5 mm and only the middle 3 mm thickness of each section was collected. Same sections from each tube were collected and then washed with DMF in an ultrasound cleaner until the agar gel was completely removed. Finally, purple precipitates were obtained after centrifugation at 12000 rpm for 10 min. The

samples from three segments were denoted as LDHNS@ZIF-67-t (top), LDHNS@ZIF-67-m (middle), and LDHNS@ZIF-67-b (bottom), respectively. For comparison, ZIF-67 samples were also prepared based on a similar method (see [Supporting Information](#)).

2.4. Preparation of CoAl-LDHNS@ZIF-67/GCE

First, a bare glassy carbon electrode (GCE, Φ = 3 mm) was polished to a mirror-like surface with 0.05 μm alumina slurry and entirely washed with ultrapure water. Second, 1 mg/mL of LDHNS@ZIF-67-m suspension was prepared by ultrasonically dispersing the hybrid in a mixed solvent of dimethyl sulfoxide (DMSO) and water (*v:v* = 1:9). 4 μL of suspension was dropped on the activated GCE surface and dried at room temperature. The modified electrode was named LDHNS@ZIF-67-m/GCE. For comparison, LDHNS@ZIF-67-t/GCE, LDHNS@ZIF-67-b/GCE, and ZIF-67/GCE were also constructed according to a similar procedure.

2.5. Sample preparation for detection

Urine samples were collected from graduate students of our group by themselves and then filtered through a 0.45 μm membrane to remove all the precipitation. 1 mL of the α -NAP and β -NAP stock solution (50 mM) was precisely diluted with the treated water to the expected concentrations. The spiked urine samples were then determined by this method under the optimal conditions.

2.6. Electrochemical tests

Electrochemical measurements were performed on a CHI660D electrochemical workstation (CHI Instruments Inc., China) using a typical three-electrode system. The modified GCE was used as a working electrode, a platinum wire as an auxiliary electrode, and a saturated calomel electrode (SCE) as a reference electrode. All measurements were carried out in a 10 mL cell containing 0.1 M phosphate buffer (PB) as a supporting electrolyte at room temperature. Before each test, the electrolyte was degassed by high-pure N_2 for at least 30 min, and the N_2 atmosphere was kept over the cell during the test. All electrodes were stored at 4 °C in a refrigerator when not in use.

3. Results and discussion

3.1. Structural and morphology characterization of materials

As the supporting matrix and exclusive Co^{2+} source for the in situ growth of ZIF-67, ultrathin CoAl-LDHNS were in advance prepared in a mixed solvent of formamide and pure water. The morphology and structure of CoAl-LDHNS and LDHNS@ZIF-67 products were firstly characterized by a scanning electron microscope (SEM) as shown in [Figs. S1 and 1](#). CoAl-LDHNS have a distinct lamellar structure with a rough surface and a thickness of about 5–10 nm. Their lateral sizes are around 100–200 nm. After the in situ growth of ZIF-67 using CoAl-LDHNS as the sole Co^{2+} source, all images of all three samples exhibit the polyhedral crystals of ZIF-67 anchored on or surrounded by the CoAl-LDHNS matrix. This observation greatly differs from the case in an unrestricted reaction system [2], in which most CoAl-LDHNS units are transformed into ZIF-67 crystals. Moreover, the slow growth kinetics in the RDF resulted in a great difference in morphology and size of ZIF-67 crystals among these three LDHNS@ZIF-67 products. The top sample of LDHNS@ZIF-67-t ([Fig. 1A](#)) displays a lot of ZIF-67 cells with the damaged and crappy dodecahedral shape and a size range of ca. 200–1000 nm, which are anchored on CoAl-LDHNS. The magnified

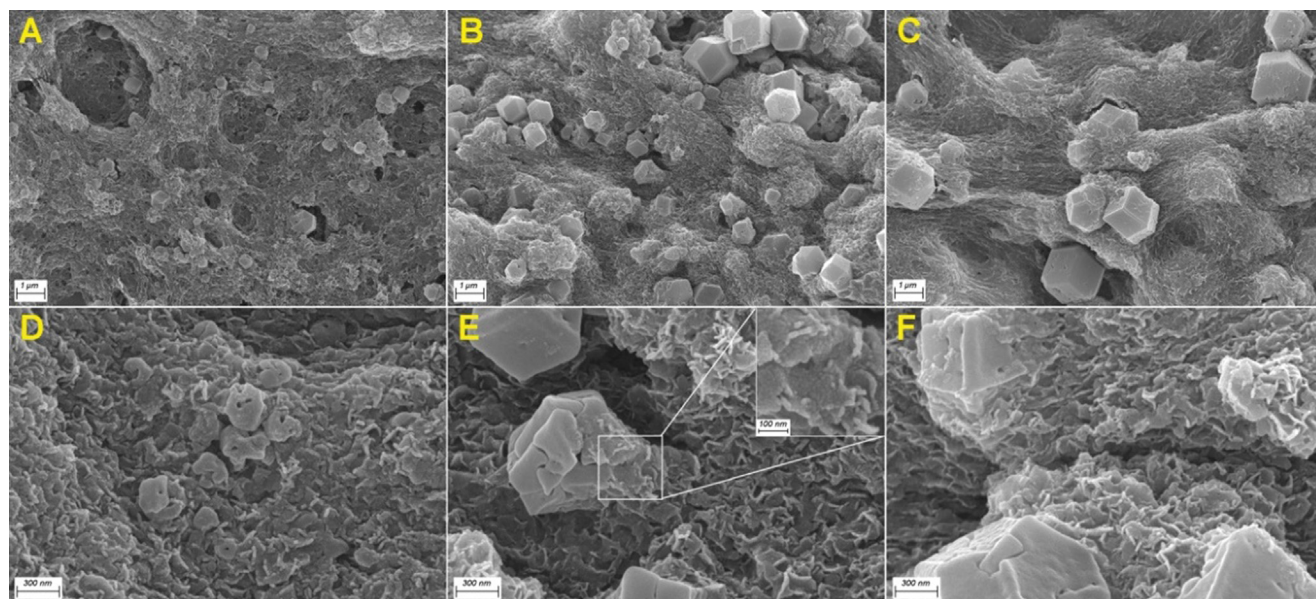


Fig. 1. SEM images of CoAl-LDHNS@ZIF-67-t (A and D), CoAl-LDHNS@ZIF-67-m (B and E), and CoAl-LDHNS@ZIF-67-b (C and F). Inset in E is the local magnified image of its selected area.

image (Fig. 1D) reveals that small ZIF-67 crystals appeared in a distorted polyhedral shape with a few holes. This phenomenon could be due to the relatively high concentration of 2-methylimidazole in the top section, which leads to a fast coordination reaction of ligands with Co^{2+} ions on the LDHNS surface and the smaller size and irregular shape of ZIF-67 crystals. As for the middle sample, LDHNS@ZIF-67-m (Fig. 1B) presents ZIF-67 crystals with a wider size range of ca. 200–2000 nm on LDHNS. This size range value is distinctly smaller than that for the ZIF-8 crystals on rGO by using the RDF technique [10], which could be ascribed to the template role of CoAl-LDHNS as the only Co^{2+} source. By contrast, despite some breakage, the polyhedral crystalline shapes of most ZIF-67 units are significantly better than those of the top sample. It can be inferred that in the middle section the relatively moderate concentration of organic ligand brings out a slow coordination reaction with CoAl-LDHNS. In this section, the nucleation process becomes gradually predominant, and thus larger particles are produced at the further diffusion distance away from the interface. The magnified image (Fig. 1E) exhibits that some crystals are composed of a few nanosheets, which is probably originated from the structural preorganization of ZIF-67 nanocrystals before forming the rhombic dodecahedral crystals [14]. The small nanocrystals can also be found on LDH plates (Inset in Fig. 1E) since some formed ZIF building blocks obstruct the diffusion of ligand molecules. While 2-methylimidazole species further vertically permeate into the bottom section, its lower concentration allows more dilatory coordination with CoAl-LDHNS than that in the middle section. Consequently, Fig. 1C displays that ZIF-67 crystals in LDHNS@ZIF-67-b have a further larger size (800–2500 nm) and a better polyhedral shape. Whereas, the amplified image (Fig. 1F) shows that some ZIF-67 crystals have a broken crystalline shape and contain several layered components. This result also reveals that the further growth of some ZIF-67 units was suppressed by this confined space in the gel matrix.

The phase compositions of samples were investigated by X-ray diffraction (XRD) patterns as shown in Fig. 2A. The ZIF-67 sample (wine curve) almost exhibits the same typical diffraction peaks with sharp shape and strong intensity between 3° and 40° [2,14] as the simulated ones from its sodalite structure (black curve). However, CoAl-LDHNS@ZIF-67 (red curve) only shows the same

characteristic diffraction lines of (003), (006), (012), (015), (018), (110), and (112) as the pure CoAl-LDH phase (blue curve) at 11.8° , 23.5° , 34.8° , 39.7° , 41.2° , 60.6° , and 61.7° , respectively [35]. Despite the visible profile of ZIF-67 in the SEM image of CoAl-LDHNS@ZIF-67 samples, its featured XRD peaks were not found in this composite. This phenomenon may be due to the low content of ZIF-67 below the XRD detection limit [36] and the surrounded ZIF-67 crystals by unreacted LDHNS (Fig. 1). To further confirm the formation of ZIF-67, the samples were also characterized by the Fourier transform infrared (FTIR) spectrum. As illustrated in Fig. 2B, CoAl-LDHNS@ZIF-67 (red curve) exhibits the distinct M–O and O–M–O vibrations respectively at about 430 and $510\text{--}620\text{ cm}^{-1}$ (overlapped with Co–N vibration of ZIF-67 at about 425 cm^{-1}) in addition to the interlayer NO_3^- vibration at about 1380 cm^{-1} , which are highly consistent with those in the LDH- NO_3^- phase (blue curve) [37]. The peaks at 1306, 1386, and 1411 cm^{-1} in the pure ZIF-67 are ascribed to the bending vibration peaks of pyrrole rings (wine curve) [38]. However, CoAl-LDHNS@ZIF-67 also presents some similar vibrations with the pure ZIF-67 phase, such as —CH— stretching vibration at about 2850 and 2930 cm^{-1} [39], C=N absorption band at about 1650 cm^{-1} , C=N vibration at around 1150 cm^{-1} , and Co—N vibration at about 750 cm^{-1} , respectively [38]. Notably, the composited catalyst displays new peaks at about 1360 cm^{-1} , which can be assigned to the combined vibrations of the interlayer NO_3^- in LDH (at about 1380 cm^{-1}), and Co—N (at about 1300 cm^{-1}) and C=N (at about 1411 cm^{-1}) in ZIF-67 [40]. These observations evidence the formation of ZIF-67 crystals on the CoAl-LDHNS matrix. The X-ray photoelectron spectroscopy (XPS) was used to characterize the elemental valence state of CoAl-LDHNS@ZIF-67. The survey spectra (the inset of Fig. 2C) identify the occurrence of Al, Co, C, N, and O on the surface of CoAl-LDHNS@ZIF-67. The Al 2p spectrum (Fig. 2C) exhibits a strong single-peak with a binding energy of 74.2 eV, which is assigned to Al^{3+} in the Al–OH form [41]. For the deconvoluted Co 2p spectrum (Fig. 2D), two peaks at 797.2 and 781.4 eV are respectively responsible for Co 2p_{1/2} and Co 2p_{3/2} with two satellite peaks at the relatively high binding energy positions, which are attributable to Co^{2+} [42]. Besides, the fine spectrum of C 1s displays three peaks of C, C—O/C—N , and O—C=O at 284.6, 286.2, and 288.3 eV, respectively (Fig. S2A), which originates from

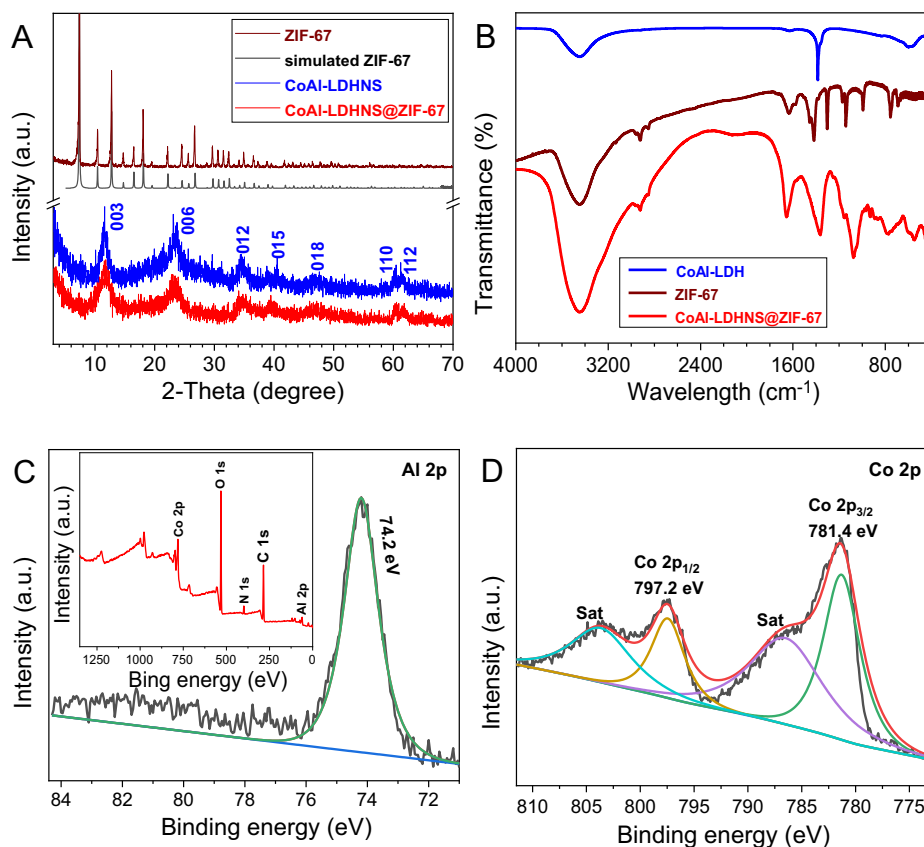


Fig. 2. (A) XRD patterns of ZIF-67, simulated ZIF-67, CoAl-LDHNS, and CoAl-LDHNS@ZIF-67. (B) IR spectra of CoAl-LDH, ZIF-67, and CoAl-LDHNS@ZIF-67. High-resolution XPS spectra of (C) Al 2p and (D) Co 2p for CoAl-LDHNS@ZIF-67. Inset in panel C is the XPS survey spectrum of CoAl-LDHNS@ZIF-67.

the presence of the interlayer CO_3^{2-} in CoAl-LDH [43]. In Fig. S2B, the N 1s spectrum reveals two peaks at 398.6 eV for pyridine-N ($\text{C}=\text{N}-\text{C}$) and 399.7 eV for pyrrolic-N ($\text{C}-\text{NH}-\text{C}$), which indexes the existence of imidazole ring in ZIF-67 [44]. The O 1s spectrum in Fig. S2C presents the bonding energy of M—OH and —OH in the LDH layer at 531.4 and 532.4 eV, respectively [45]. Based on the above analysis, it is established that CoAl-LDHNS@ZIF-67 was successfully prepared by the RDF technique.

3.2. Electrochemical characterization of modified electrodes

The electrochemical performances of different electrodes were studied by cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS). Fig. 3A presents their CV curves in a 5.0 mM $\text{Fe}(\text{CN})_6^{3-/4-}$ (1:1) solution containing 0.1 M KCl. Compared to bare GCE, CoAl-LDHNS/GCE (curve b) and ZIF-67/GCE (curve c) display stronger electrochemical responses owing to their plentiful catalytic sites and large surface area. However, both of them give rise to bigger peak-to-peak separations (ΔE_p : 142 and 121 mV for CoAl-LDHNS and ZIF-67, respectively) than that of bare GCE (101 mV) because of their inherently non-conductive properties. Impressively, CoAl-LDHNS@ZIF-67/GCE yields the largely reduced ΔE_p value of 108 mV and the significantly increased redox peak currents with better reversibility ($I_{pa} = 131.2 \mu\text{A}$ and $I_{pc} = 127.6 \mu\text{A}$, $I_{pa}/I_{pc} = 1.028$). Based on CV curves in Fig. 3a, the electrochemically active surface areas (EASA) of CoAl-LDHNS/GCE, ZIF-67/GCE, and CoAl-LDHNS@ZIF-67/GCE were respectively calculated to be 0.39 cm^2 , 0.45 cm^2 , and 0.54 cm^2 according to the Randles-Sevcik equation [46,47]. It is distinct that CoAl-LDHNS@ZIF-67 gained a larger EASA value than the CoAl-LDHNS/GCE or ZIF-67/GCE reference. The above results indicate the improved electron transfer

rate and electrocatalytic kinetics on CoAl-LDHNS@ZIF-67/GCE. The charge transport property of modified electrodes was also evaluated by electrical impedance spectroscopy (EIS) technique using $\text{Fe}(\text{CN})_6^{3-/4-}$ as the redox probe. Fig. 3B displays Nyquist plots of GCE, CoAl-LDHNS/GCE, ZIF-67/GCE, and CoAl-LDHNS@ZIF-67/GCE with the corresponding Randles' equivalent circuit based on the impedance equations (Inset of Fig. 3B). R_Ω and R_{ct} stand for the resistance of electrolyte and the charge-transfer resistance, respectively. The almost same intercepts of the X-axis (R_Ω) for all electrodes index their equivalent mass-transfer resistance. Nevertheless, their semicircular diameter values (R_{ct}) follow a similar sequence to that of the CV data. CoAl-LDHNS/GCE (curve b) and ZIF-67/GCE (curve c) display smaller charge resistances (R_{ct} : 645 and 434 Ω , respectively) than that of the bare GCE (curve a, R_{ct} : 811 Ω). It is suggested that, despite poor conductivity, ultrathin CoAl-LDHNS and porous ZIF-67 provide more catalytic centers and available mass-transfer channels. The CoAl-LDHNS@ZIF-67 composite (curve d) delivers the smallest R_{ct} value of 114 Ω , suggesting the further lowered interface resistance of electron transfer. Based on the CV and EIS results, in the growth of ZIF-67 on the CoAl-LDHNS process these two components mutually inhibited the aggregation of both of them, thus ensuring more accessible catalytic sites and channels for easier electron communication. To assess the effect of vertical gradient on products, the CVs and Nyquist plots of CoAl-LDHNS@ZIF-67-t/GCE (a), CoAl-LDHNS@ZIF-67-m/GCE (b), and CoAl-LDHNS@ZIF-67-b/GCE (c) were determined (Fig. S3). The middle sample demonstrates the largest redox signals and smallest charge-transfer resistance, which could be attributed to the appropriate ZIF-67 crystal size and size range with good dispersity on the LDHNS matrix. For con-

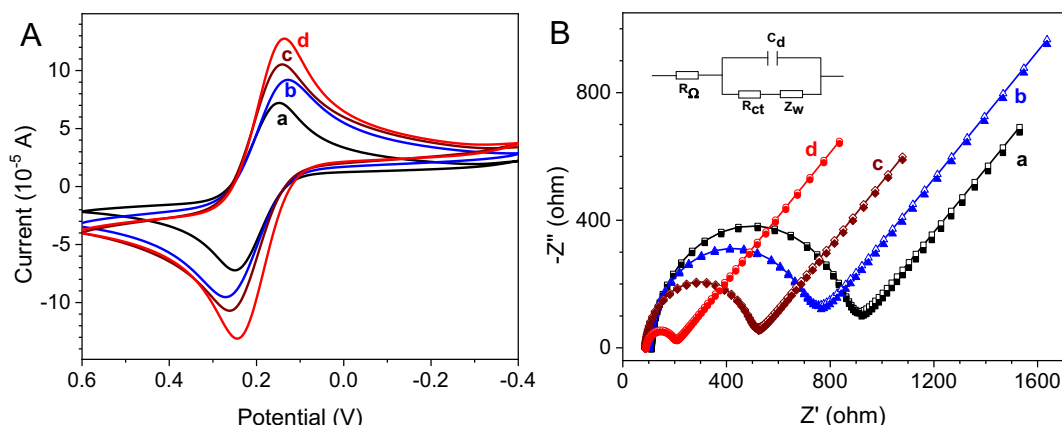


Fig. 3. (A) CVs and (B) Nyquist plots of bare GCE (a), CoAl-LDHNS/GCE (b), ZIF-67/GCE (c), and CoAl-LDHNS@ZIF-67/GCE (d) in 5.0 mM $\text{Fe}(\text{CN})_6^{3-/4-}$ (1:1) solution containing 0.1 M KCl (scan rate: 100 mV/s). In Nyquist plots, measured data were shown as solid symbols with calculated fitting to the equivalent circuit as hollow symbols with the fitting line. Inset is the Randles circuit.

venient understanding, the following CoAl-LDHNS@ZIF-67 product represents the middle sample unless otherwise noted.

3.3. Electrochemical behaviors of α -NAP and β -NAP on modified electrodes

Fig. 4A depicts the CV curves of CoAl-LDHNS@ZIF-67/GCE in the absence of NAP and presence of α -NAP, β -NAP, and α -NAP + β -NAP in PB (pH = 7.5). As expected, both α -NAP and β -NAP in the mixture exhibit two separated anodic peaks at around 0.33 and 0.49 V with the similar shape, and nearby equal peak potential and current to those of α -NAP and β -NAP alone, revealing only the electrochemical reaction of NAP isomers. The peak separation of α -NAP and β -NAP is estimated to be about 162 mV, which is wide enough for their simultaneous determination. Fig. 4B displays the voltammetric behaviors of NAP isomers on different electrodes. The bare GCE shows the lowest electrochemical responses of NAP isomers with a peak potential separation of 159 mV (curve a). Compared to the bare GCE, the modification of CoAl-LDHNS (curve b) or ZIF-67 (curve c) both enhanced the anodic signals of α -NAP and β -NAP to some extent. The CoAl-LDHNS@ZIF-67/GCE (curve d) offers the biggest anodic peak currents and peak potential difference, suggesting that the in situ growth of ZIF-67 on ultrathin the LDHNS produces more active sites and larger surface area for faster electron transfer and more efficient catalysis than the single counterparts of CoAl-LDHNS or ZIF-67. For all electrodes, the absence of

any reduction signal of NAP isomers reveals the irreversibility of NAP molecule oxidation. The possible electrocatalytic mechanism can be summarized as the following. On the CoAl-LDHNS@ZIF-67 surface, abundant reactive sites of N-atom and hydroxyls are favorable to form the hydrogen bond with hydroxyl groups of NAP molecules. The presence of ZIF-67 can further facilitate the entrapment of NAP isomers through the π - π and hydrophobic interaction between the imidazole ring and the aromatic ring. Moreover, the inorganic LDHNS provide the more available channels for fast and convenient mass transfer and charge transfer. It should be noted that the CoAl-LDHNS and ZIF-67 blocks are rationally assembled with a certain ratio by the RDF technique, thus CoAl-LDHNS@ZIF-67/GCE gives rise to the strongest electrocatalytic signals by integrating the capturing and electrocatalytic ability of these two phases toward NAP isomers. The oxidation process of NAP isomers can be expected since the O—H bond in NAP molecules can be weakened and are more easily broken under the force of the hydrogen bond between —OH groups in NAP and LDH. On the other side, due to the different charge densities, the oxidation of hydroxyl groups in α -NAP and β -NAP exhibits different anodic potentials. It is known that the p- π conjugation between —OH groups and naphthalene skeleton can lower the electron cloud density of the O-atom and make the polarity of the O—H bond stronger, thus improving its ionization activity. Given the bigger electron cloud density of —OH at α -position, α -NAP is more easily oxidized with a much smaller oxidation potential relative to β -

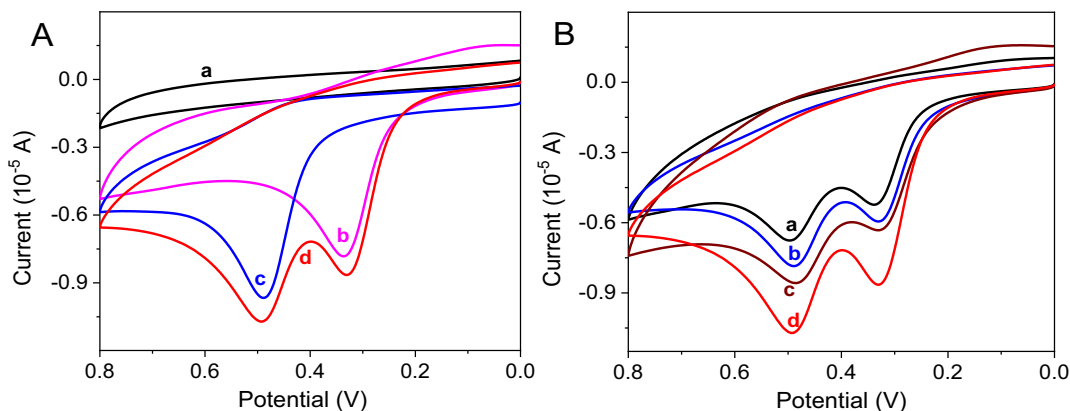


Fig. 4. (A) CVs of CoAl-LDHNS@ZIF-67/GCE in blank (a), α -NAP (b), β -NAP (c), and α -NAP + β -NAP (d) buffer. (B) CVs of α -NAP + β -NAP on bare GCE (a), CoAl-LDHNS/GCE (b), ZIF-67/GCE (c), and CoAl-LDHNS@ZIF-67/GCE (d) in PB (pH = 7.5). α -NAP: 0.1 mM, β -NAP: 0.1 mM, α -NAP + β -NAP: 0.1 mM. Scan rate is 100 mVs⁻¹.

NAP. Therefore, it is reasonable for the simultaneous detection of α -NAP and β -NAP. Fig. S4 displays the CV curves of NAP isomers at the three modifiers obtained from the different gradient depths. The middle sample of CoAl-LDHNS@ZIF-67 yields the best electrocatalytic performance toward α -NAP and β -NAP. This result is consistent with those observed in Fig. S3.

3.4. Influences of CoAl-LDHNS@ZIF-67 content, accumulation time, and potential

The major parameters including modifier loading, accumulation time, and accumulation potential can influence the performance of the modified electrode. Therefore, different loadings of CoAl-LDHNS@ZIF-67 suspension were firstly investigated by CV in PB (pH 7.5) containing NAP isomers. As shown in Fig. S5A, the oxidation peak currents of both NAP isomers reach the summit at 4 μ L in the selected range, implying that less and excess sensing material ink are both not able to perform best due to the formation of a thinner or thicker film. Fig. S5B displays the influence of accumulation time. The oxidative peak currents of α -NAP and β -NAP initially increase against accumulation time from 20 to 60 s, and then decrease from that moment. This observation means that the adsorption of NAP isomers on the LDHNS@ZIF-8 film achieves saturation at 60 s. The result of the accumulation potential demonstrates that their anodic peak currents come to the maximum at -0.2 V (Fig. S5C). As a result, the optimized conditions are obtained as 4 μ L, 60 s, and -0.2 V for the loadings, the accumulation time, and accumulation potential, respectively.

3.5. Influences of pH and scan rate

The influence of PB with different pH on the anodic peak potential and current (E_{pa} and I_{pa}) of NAP isomers (both concentrations are 0.1 mM) were analyzed by CV on CoAl-LDHNS@ZIF-67/GCE at a scan rate of 100 mV s^{-1} (Fig. 5A). Panel B in Fig. 5 shows that both E_{pa} values of α -NAP and β -NAP linearly decrease while increasing pH from 6.0 to 8.0, indicating proton taking part in the electrode reaction [48]. The good linear relationships of E_{pa} vs pH values can be expressed by the following two equations: $E_{pa}(\alpha\text{-NAP})(\text{V}) = -0.059 \text{ pH} + 0.7904$ ($R^2 = 0.997$) and $E_{pa}(\beta\text{-NAP})(\text{V}) = -0.064 \text{ pH} + 0.9982$ ($R^2 = 0.999$), respectively. In line with the Nernst equation: $E_{pa} = E^0 - (0.059 \text{ m/n})\text{pH}$, the slopes of both regression equations are close to the theoretical value (-0.059), revealing an electro-oxidation process of NAP isomers concerning the equal number of protons (m) and electrons (n) [49]. Additionally, the I_{pa} values of α -NAP and β -NAP are maximum at the pH of

7.5 in a range from 6.0 to 8.0. Therefore, a PB with a pH of 7.5 was chosen as the supporting electrolyte in all related experiments.

The influence of the scan rate on I_{pa} of NAP was also investigated by CV. Fig. 6A displays CV curves of NAP isomers (0.1 mM) on CoAl-LDHNS@ZIF-67/GCE in PB (pH 7.5, 0.1 M) with various scan rates from 20 to 160 mV s^{-1} . Fig. 6B exhibits that both I_{pa} values proportionally increase with the increase of scan rate in the given range, revealing a typical adsorption controlled process for the oxidation of NAP isomers on the modified electrode [2,50]. On the other hand, both E_{pa} values linearly increase as the increase of natural logarithm of scan rate ($\ln v$) as shown in Fig. 6C. Their good linearity can be expressed by the following two equations: $E_{pa}(\alpha\text{-NAP})(\text{V}) = 0.0189 \ln v + 0.2691$ ($R^2 = 0.978$) and $E_{pa}(\beta\text{-NAP})(\text{V}) = 0.0190 \ln v + 0.4422$ ($R^2 = 0.953$), respectively. Generally, E_{pa} for a fully irreversible electrode reaction can be expressed by the following transferred Laviron equation [51]:

where E^0 is the formal potential; D is the diffusion coefficient; v is scan rate; α is transfer coefficient (assumed as 0.5 for the completely irreversible electrode process) [2]; n is electron transfer number; k_0 is the standard rate constant; and R , T , and F are respectively gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), temperature (298 K), and Faraday constant (96485 C mol^{-1}). Based on the above regression equations and the Laviron equation, n values for α -NAP and β -NAP are respectively calculated to be 1.36 and 1.35. Both values are approaching one, meaning the one-proton accompanying one-electron pathway for NAP isomers oxidation. The oxidation of NAP isomers in this work can thus be illustrated as Fig. 7 [1,4].

3.6. Simultaneous differential pulse voltammetric determination

Under the optimal conditions, the simultaneous detection of α -NAP and β -NAP was conducted on CoAl-LDHNS@ZIF-67/GCE by differential pulse voltammetry (DPV) as shown in Fig. 8A. Fig. 8B displays plots of the I_{pa} values upon their concentrations. Both plots exhibit good linearity in two different sections. Therein, α -NAP gives rise to two linear portions from 0.2 to 50 μM and 50 to 120 μM with regression equations of $I_{pa}(\mu\text{A}) = -0.0317 C_{\alpha\text{-NAP}}(\mu\text{M}) - 0.673$ ($R^2 = 0.986$) and $I_{pa}(\mu\text{A}) = -0.0221 C_{\alpha\text{-NAP}}(\mu\text{M}) - 1.224$ ($R^2 = 0.963$), respectively. Whereas β -NAP demonstrates two linear portions from 0.2 to 60 μM and 60 to 120 μM with regression equations of $I_{pa}(\mu\text{A}) = -0.0253 C_{\beta\text{-NAP}}(\mu\text{M}) - 0.848$ ($R^2 = 0.988$) and $I_{pa}(\mu\text{A}) = -0.0085 C_{\beta\text{-NAP}}(\mu\text{M}) - 2.019$ ($R^2 = 0.971$), respectively. Their limits of detection (LOD) ($S/N = 3:1$) were obtained to be 54 and 82 nM for α -NAP and β -NAP, respectively. Table 1 summarizes the analytical performance of recently reported electrochemical methods and this work. The LOD values of this work are larger than those of rGO@ZIF-8/GCE,

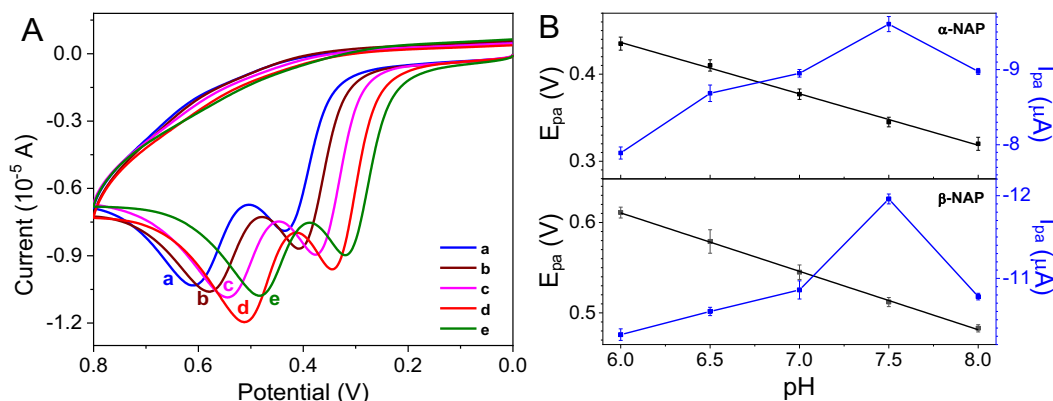


Fig. 5. CVs (A) of 0.1 mM α -NAP and β -NAP mixture at CoAl-LDHNS@ZIF-67/GCE in 0.1 M PB with different pH values (from a to f: 6.0, 6.5, 7.0, 7.5, and 8.0). Plots of oxidation peak potential (E_{pa}) and peak current (I_{pa}) vs. pH value. The scan rate is 100 mV s^{-1} .

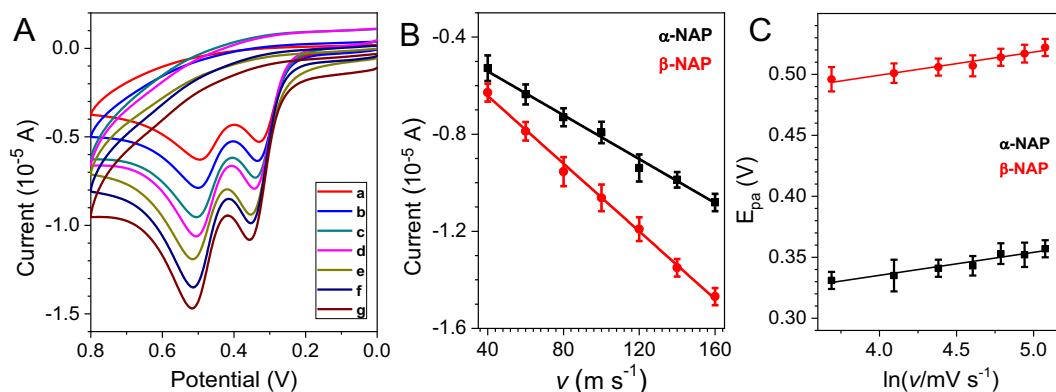


Fig. 6. (A) CVs of α -NAP and β -NAP mixture (0.1 mM) on CoAl-LDHNS@ZIF-67/GCE in PB (0.1 M, pH = 7.5) at different scan rates (a-g: 40, 60, 80, 100, 120, 140, and 160 mV s^{-1}). (b) The linear relationship of I_{pa} against scan rate. (c) The linear relationship of E_{pa} and $\ln v$.

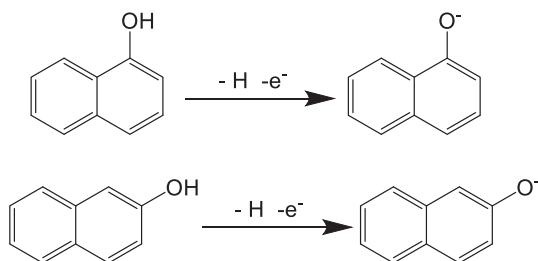


Fig. 7. The oxidation mechanism of α -NAP and β -NAP on CoAl-LDHNS@ZIF-67/GCE.

which could probably be attributed to the absence of a conductive component in CoAl-LDHNS@ZIF-67 compared to the rGO@ZIF-8 modifier [10]. However, the analytical parameters are superior or comparable to those of the other sensors in Table 1.

3.7. Reproducibility, stability, and interference

Five CoAl-LDHNS@ZIF-67/GCE electrodes were prepared in the same procedure and were employed for the assessment of their reproducibility and stability. The oxidative peak currents of α -NAP and β -NAP mixture (20 μM) on five parallel modified electrodes produce an acceptable reproducibility with a relative standard deviation (RSD) of 4.56%. The anodic peak signals of NAP isomers retain more than 90% of their initial signals after three-week storage at 4 $^{\circ}\text{C}$, proving good stability of the modified electrode. The interference experiment was conducted by determining

the amperometric response of different phenolic analogues, such as 4-aminophenol, phenol, hydroxyphenol, phloroglucinol, and bisphenol A. As shown in Fig. 9A and B, despite a little bigger response of 4-aminophenol than those of the other four phenols, the above five phenolic compounds gave rise to insignificant signals compared to those of α -NAP and β -NAP. Furthermore, we also determined DPV curves of NAP isomers and the five phenolic interfering substances with the same concentration. As illustrated in Fig. 9C, phenol, hydroxyphenol, phloroglucinol, and bisphenol A display either far peak potentials away from NAP isomers or/and much smaller oxidation currents, consequently generating the negligible influence. Nevertheless, the oxidation peak potential of 4-aminophenol is close to that of α -NAP with a bigger peak signal. Hence, 4-aminophenol shows relatively more obvious interference than the other four interferents. This result strongly supported the amperometric data. It thus confirmed the good selectivity of this method for the simultaneous detection of NAP isomers.

3.8. Practical application

To estimate the practical applicability, CoAl-LDHNS@ZIF-67/GCE was used for the simultaneous detection of α -NAP and β -NAP in the human urine samples. The known amounts of α -NAP and β -NAP were spiked in the urine samples. The determined DPV curves are shown in Fig. S6 and the related results are outlined in Table S1, in which the recoveries of the two samples are ranged from 98.7% to 101.3% with RSD lower than 5%. The analytical results from high performance liquid chromatography (HPLC) are

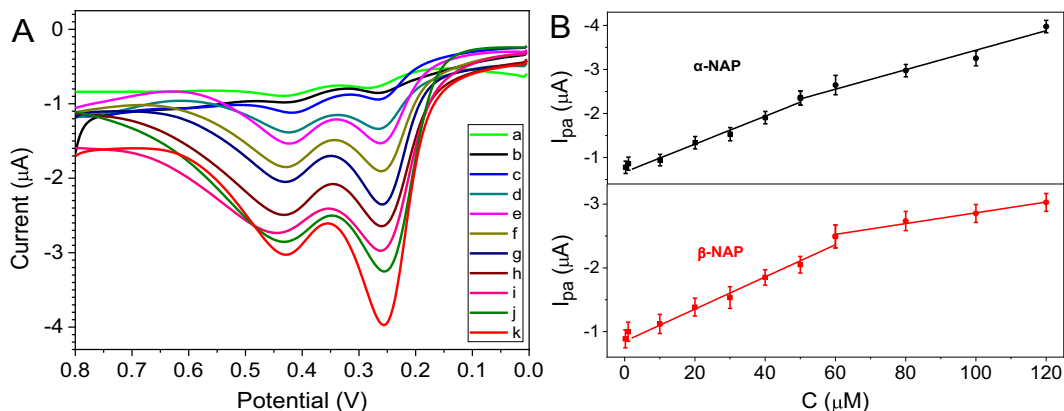


Fig. 8. (A) DPV curves of the mixture containing α -NAP and β -NAP in different concentrations (a-k: 0.2, 1, 10, 20, 30, 40, 50, 60, 80, 100, and 120 μM) at CoAl-LDHNS@ZIF-67/GCE. (B) The responding calibration curves of peak currents against concentrations of α -NAP and β -NAP.

Table 1Comparison of this sensor for the determination of α -NAP and β -NAP with others.

Modified electrodes	Analytical technique	Linear range (μM)		LOD (nM)		Ref.
		α -NAP	β -NAP	α -NAP	β -NAP	
CPT-BDD	DPV	0.2–3.85	0.2–3.85	50	100	[52]
β -CD/TiO ₂ /QCM	QCM	1–130	1–120	836	215	[53]
PAO film/GCE	DPV	0.2–3.2	–	800	200	[4]
P3MT-nano-Au	SDV	0.7–150	1.0–150	100	300	[54]
LDHNS@ZIF-67/GCE	DPV	0.3–150	0.3–150	62	94	[2]
PtNPs/CNTs/GCE	LSV	1–800	1–800	500	600	[55]
rGO@ZIF-8/GCE	Amperometry	0.05–12	0.02–15	15	17	[10]
CoAl-LDHNS@ZIF-67/GCE	DPV	0.3–150	0.3–150	54	82	this work

Note: CPT-BDD: cathodically pretreated boron-doped diamond; β -CD: β -cyclodextrin; P3MT: poly(3-methylthiophene); PtNPs/CNTs: Pt nanoparticles/carbon nanotubes; PAO: poly (acridine orange); QCM: quartz crystal microbalance; SDV: semi-derivative voltammograms; LSV: linear sweep voltammetry.

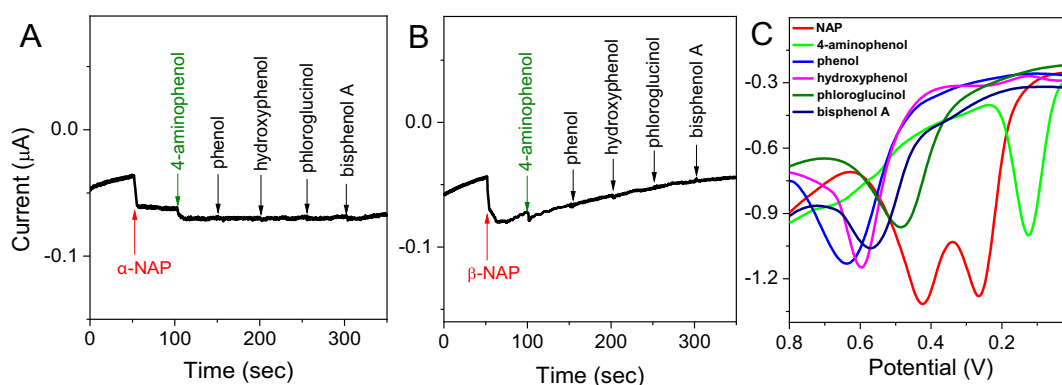


Fig. 9. Amperometric curves of (A) α -NAP (10 μM) and (B) β -NAP (10 μM) on CoAl-LDHNS@ZIF-67/GCE while adding 10 μM different interfering species in 0.1 M PB (pH 7.5) at the applied potentials of 0.26 and 0.42 V, respectively. (C) DPV curves of common phenolic compounds and α -NAP (20 μM) and β -NAP (20 μM) at CoAl-LDHNS@ZIF-67/GCE in the 0.1 M PB (pH 7.5).

also obtained and the recovery range is from 99.2% to 103.1%. The results indicate the good accuracy and applicability of this method.

4. Conclusion

Compared to an open reaction environment [2], a reaction-diffusion framework (RDF) can control the coordination reaction kinetics because of its confined space. Herein, the RDF technique was employed for the in situ growth of zeolitic imidazole framework-67 (ZIF-67) on the ultrathin CoAl-layered double hydroxide nanosheets (CoAl-LDHNS) in the gel without an additional Co^{2+} source. Owing to the natural vertical diffusion of 2-methyl imidazole in the agar gel matrix, the coordination reaction between the ligands and CoAl-LDHNS units results in the controllable growth of ZIF-67 nanocrystals in the restricted space. By contrast with the transformation of most CoAl-LDHNS units into ZIF-67 [2], the CoAl-LDHNS@ZIF-67 composite here occurs in a certain ratio of LDHNS to ZIF and offers a large surface area and ample catalytic sites. The obtained CoAl-LDHNS@ZIF-67/GCE exhibits good catalytic performance for the simultaneous detection of naphthol isomers with excellent selectivity, stability, and reproducibility. The limits of detection of this method are much lower than those of several reported methods [52–55]. This work provides some inspiration for the fabrication of an electrochemical sensing platform by combining ultrathin LDHNS and ZIF building blocks. However, the thickness and the content of LDHNS in the gel matrix are essential for the yield, morphology, and size distribution of ZIF crystals, which are significant for the catalytic activity and need to be investigated in the future.

CRediT authorship contribution statement

Ying Liu: Data curation, Software, Writing - original draft. **Runxia Wang:** Data curation, Software, Writing - original draft. **Haiyan Wang:** Data curation, Software, Investigation. **Xijuan Yu:** Supervision. **Xien Liu:** Writing - review & editing. **Peng He:** Supervision. **Xilin She:** Supervision. **Tianrong Zhan:** Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We are grateful to the Natural Science Foundation of Shandong Province, China (No. ZR2019MB062, ZR2014JL013), the Key Research and Development Program of Shandong Province (2017GGX20143), Taishan Scholar Program of Shandong Province of China (No. ts201712045), the Foundation of Key Laboratory of Sensor Analysis of Tumor Marker, Ministry of Education, QUST (SATM201603), the foundation of Key Laboratory of Colloid and Interface Chemistry (Shandong University), Ministry of Education (201702), the open foundation from the Key Lab of Marine Bioactive Substance and Modern Analysis Technology, SOA (MBSMAT-2017-02, MBSMAT-2016-02, and MBSMAT-2015-04).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2021.04.141>.

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